Project Demo Planning: Structured Demonstration Script

Project: NeuroClass — Multimodal Attendance & Engagement Agent

Module: `8.Project Demonstration`

Phase: Demonstration Planning & Walkthrough Script (Solo Developer Project)

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1. Demonstration Overview & Setup Checklist

This document provides a structured, timing-calibrated demonstration script for presenting the NeuroClass system to evaluators, faculty review committees, and technical stakeholders.

Pre-Demo Checklist (Solo Presenter)

- [ ] Terminal 1: Active virtual environment with FastAPI backend running (`python run_from_root.py --agent`).
- [ ] Terminal 2: Streamlit companion dashboard running (`python run_from_root.py --streamlit`).
- [ ] Browser Window 1: Opened to `http://127.0.0.1:8000` (FastAPI Web HUD).
- [ ] Browser Window 2: Opened to `http://127.0.0.1:8501` (Streamlit Analytics Cockpit).
- [ ] Camera & Microphone: Verified accessible and functional.
- [ ] Sample Video / Simulated Mode: Ready as fallback if physical webcam is occupied.

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2. Walkthrough Demonstration Script (12 Minutes)

DEMO TIMELINE & NARRATIVE STRUCTURE			
Time	Demo Segment	On-Screen Actions & Commands	Spoken Narrative
02:00	1. Environment &	Run `python validate_environment.py`	"100% offline,
	Privacy Audit	Highlight: Zero cloud network egress	zero cloud STT"
03:00	2. 3-Way Hybrid	Open HUD (`http://127.0.0.1:8000`). Show track.	"Effortless
	Enrollment Demo	Click `[IDENTIFY / LINK] Identify / Link` -> Link to `STU-109`.	zero-delay enroll
02:30	3. Scenario 1 Demo	Walk across camera (<15s) -> Discarded.	"Walk-bys filtered
	Sustained Att.	Remain seated -> Presence counter hits 65%.	verified presence
02:30	4. Scenario 2 Demo	Turn head left/right (>25°) -> Attentive drops.	"Multimodal gaze
	Multimodal Fuse	Speak into mic -> RMS & VAD bars react live.	& voice dynamics"
02:00	5. Scenario 3 Demo	Speak continuously -> Monologue toast triggers.	"Real-time AI
	AI Teaching Nudge	Trigger 2nd event -> Cooldown blocks spam.	teaching copilot"
01:30	6. Analytics &	Open Streamlit (`http://127.0.0.1:8501`).	"Tamper-evident
	CSV Export Demo	Show focus heatmaps & export verified CSV.	audit compliance"

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3. Step-by-Step Segment Execution Guide

The demonstration is partitioned into four distinct chronological segments designed to showcase seamless system setup, automated attendance tracking, live engagement nudges, and post-session CSV export.

Segment 1: Environment & Privacy Verification (Minutes 0:00 - 2:00)

- **Action:** Open terminal and run:

```
python validate_environment.py
```

- **Key Talking Points:**

- *"NeuroClass operates with certified zero external network dependency. No student faces or voice streams are ever transmitted to cloud servers."*
- *"All 14 core dependencies (OpenCV, Librosa, MediaPipe, NumPy, FastAPI) are isolated in a local virtual environment."*

Segment 2: Live Ingestion & 3-Way Hybrid Enrollment (Minutes 2:00 - 5:00)

- **Action:** Show the Web HUD at `http://127.0.0.1:8000`. Point out the active video feed with facial privacy blurring enabled.
- **Action:** Point out an unidentified participant marked as `TRK-001`.
- **Action:** Click `[IDENTIFY / LINK] Identify / Link`, enter `STU-109` (*Ada Lovelace*), and click submit.
- **Key Talking Points:**
- *"Traditional biometric systems fail when a student is not pre-registered. NeuroClass solves this with 3-Way Hybrid Enrollment: file drop, web upload, or live 1-click linking directly from the video HUD."*
- *"The system instantly creates* `student_db/STU-109.jpg` *and links all historical presence records retroactively without losing a single frame."*

Segment 3: Scenario 1 — Sustained Presence & Walk-By Filter (Minutes 5:00 - 7:30)

- **Action:** Move quickly across the frame for 6 seconds and leave. Show that the walk-by filter records no verified attendance mark.
- **Action:** Return to seat and remain in frame. Observe the active presence percentage rise until crossing the ***65\%*** threshold.
- **Key Talking Points:**
- *"A student walking past an open door is detected for* ***<15****s and discarded as non-participant noise."*
- *"Verified attendance requires sustained presence across* ***\ge 65\%*** *of the observation window, eliminating proxy check-ins."*

Segment 4: Scenario 2 — Multimodal Temporal Fusion (Minutes 7:30 - 10:00)

- **Action:** Look directly forward at the camera. The Visual Attention HUD displays `95% Attentive (Yaw: +2°, Pitch: -4°)`.
- **Action:** Turn head sharply to the side (***>25^\circ***). Attention drops to `Distracted (Yaw: 34°)`.
- **Action:** Speak into the microphone. Point to the Librosa Acoustic HUD showing active RMS and Voice Activity Detection (VAD).
- **Key Talking Points:**
- *"The Multimodal Fusion Engine unifies visual gaze (****50\%****), motion balance (****20\%****), and acoustic dynamics (****30\%****) into a single, explainable 0-100% Engagement Index."*

Segment 5: Scenario 3 — Autonomous Pedagogical Nudges (Minutes 10:00 - 12:00)

- **Action:** Maintain continuous speech for simulated duration or simulate drop. Observe the toast notification appear on the top-right of the HUD:

> *"Pacing Alert: Continuous monologue detected. Consider pausing for an interactive poll."*

- **Action:** Show that the system locks out redundant notifications for 45 seconds to prevent instructor distraction.
- **Action:** Conclude session and show the Streamlit Analytics dashboard (`http://127.0.0.1:8501`), downloading the official CSV attendance roster.